

Technology Stack

Date	27 june 2026
Team ID	xxxxxx
Project Name	Opticrop-Smart Agricultural Production Optimization Engine”
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript	Ensures a simple, intuitive, and mobile-responsive user interface. This is critical for farmers who may be accessing the application via mobile devices while out in the field.

architecture and provide a brief justification for your choice.

Technology Stack Details

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask Web Framework), Scikit-learn, Pandas, NumPy, SciPy	Flask provides a lightweight, fast web server, while Python natively supports the core machine learning and data processing libraries required for the optimization engine to process agricultural data.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	PostgreSQL or MySQL	provides strong relational data mapping, which is ideal for securely structuring and storing environmental parameters, user login credentials, and historical crop-yield datasets.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Heroku or AWS (Amazon Web Services)	Offers easy scalability and cost-effective hosting. This ensures the application remains highly available and can handle surges in traffic during peak planting seasons.

5	Version Control & CI/CD	<i>GitHub</i>	Enables seamless collaboration, code sharing, and version tracking among all team members during the development of the ML models and web application.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	OpenWeatherMap API (or similar Weather API)	<i>enables the system to automatically fetch and integrate real-time climate data (such as temperature, humidity, and rainfall) if the user does not manually input it for their assessment.]</i>
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		